

Stage 4.3 - Final Methodology and Results

Targeted local-route analysis and full-discovery bidirectional extension for OLMo-2-1124-7B

25 September 2026

Revision status. This document supersedes the earlier S4.3 initial-panel-only analysis report and replaces the prospective wording for S4.3 in the 24 September Stage 4 report. It does not alter the already executed S4.1/S4.2 results.

1. Scope and stopping boundary

Stage 4.3 was designed as a bounded follow-up to S4.1/S4.2. It tested a fixed local registry and two proposed composed chains, then extended a prospectively selected subset across the full 89-family discovery population in both noising and reverse-restoration directions. The broad all-head/MLP search, automatic second round, fallback block scan, and automatic within-block head decomposition remained excluded.

S4.3 is now complete under that plan. The held-out families were not accessed. The results below therefore support claims about the tested discovery population and interventions, not held-out generalization, circuit uniqueness, or a complete semantic mechanism.

2. Fixed model, population, and causal quantity

- Model: allenai/OLMo-2-1124-7B.
- Pinned revision: 7df9a82518afdecae4e8c026b27adccc8c1f0032.
- Discovery population: 89 families x 2 prompt orders = 178 prompt pairs.
- Initial S4.3 panel: 20 common families x 2 orders = 40 prompt pairs.
- Held-out population: sealed throughout S4.3.
- Main behavioral quantity: clean-minus-corrupt answer-logit margin at the first divergent token, with a teacher-forced shared answer prefix where required.
- Noising effect I: $M(\text{clean recipient}) - M(\text{clean recipient with corrupt donor route})$.
- Reverse-restoration effect J: $M(\text{corrupt recipient with clean donor route}) - M(\text{corrupt recipient})$.

3. Intervention semantics and matched comparator

The local-route engine starts from a source-replaced hybrid while freezing undeclared downstream residual branch increments after their normalization. Only the named MLPs/blocks or intermediate head outputs are released. The receiver's actual Q/K/V is then recaptured under that hybrid and the selected channel is injected into a fresh recipient. Whole-block releases are interpreted only as block-mediated interventions.

For every selected non-direct route, the analysis uses the prospectively matched direct/no-release comparator with the same source, source site, receiver, and channel. The screened matched increment is:

$$\text{Delta} = \text{raw route effect} - \text{matched direct/no-release effect}$$

This subtraction is performed within each pair, prompt order, and intervention direction before the two prompt orders are averaged to one family-level observation.

4. Frozen descriptive retention rule

- Threshold $\tau = 0.10$ logits.
- Coherent: absolute signed family mean ≥ 0.10 and at least 70% of families agree with the mean sign.
- Heterogeneous: mean absolute family effect ≥ 0.10 and at least 20% of families have absolute effect ≥ 0.10 .
- Retained: coherent OR heterogeneous.
- The rule is descriptive; it is not a significance test.

For the full extension, the same rule is applied separately to noising and reverse restoration. No configuration is added, removed, or re-ranked after seeing the full-discovery results.

5. Initial 40-pair screen and frozen extension set

The initial noising screen evaluated the full 305-configuration S4.3 registry on 40 prompt pairs (20 families x 2 orders), producing 12,200 scientific records. Among 220 non-direct route configurations, 22 met the frozen retention rule; 20 were coherent and heterogeneous, and 2 were heterogeneous only. All 24 L13H18 alternate-slot controls were non-retained.

Source	Tested non-direct	Retained	Selected for extension
L13H18	96	0	0
L8H15	48	4	4
L16H31	20	0	0
L20H7	20	2	1
L26H23	12	3	2
L15H25	12	5	3
L17H1	12	8	6

The predeclared extension cap selected 16 retained contrasts plus 11 distinct matched direct prerequisites. Six other initially retained configurations were omitted only because of the cap; they remain initial-panel results and were not upgraded by the full-discovery extension.

6. Production provenance and compatibility amendment

The initial-v2 40-pair production used the Stage-4 margin for its historical compatibility check after a baseline-only diagnostic showed that individual saved logits could differ slightly while the margin remained within tolerance. The completed initial-v2 run produced 12,200/12,200 records.

Before the 178-pair extension, the original global historical-margin tolerance of 0.05 failed deterministically for the two orders of family 034 on both A100-SXM4-40GB and A100-SXM4-80GB. A provenance audit authenticated the original Stage-3 artifact and verified that every inherited family-034 field, including the historical clean logits and prompts, was unchanged in the S4.3 frozen input. The only added fields were derived S4.3 execution metadata.

The final amendment therefore did not relax the global threshold. The other 176 pairs still had to satisfy historical margin error ≤ 0.05 . Exactly the two family-034 pair IDs instead had to reproduce a separately frozen baseline diagnostic reference within $1e-6$. The maximum reference error was 0.0. No extension scientific record was produced before this amendment was frozen.

Compatibility handling is a provenance/implementation accommodation, not a scientific exclusion: both family-034 pairs remain in the 178-pair scientific population.
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7. Full-discovery bidirectional extension

The final production run evaluated exactly the frozen 16 selected contrasts and 11 prerequisites on all 178 discovery pairs in both noising and reverse restoration. It produced 9,612 scientific records: 4,806

noising and 4,806 restoration. The full-178 compatibility gate, bidirectional self-donor gate, and all atomic pair checkpoints passed; held-out remained sealed.

Item	Final status
Prompt pairs	178 / 178
Discovery families	89 / 89
Configs / direction	27 (16 selected + 11 prerequisites)
Scientific records	9,612
Noising / restoration	4,806 / 4,806
Selected routes retained	16/16 in noising; 16/16 in restoration
Coherent	14/16 in noising; 14/16 in restoration
Heterogeneous	16/16 in noising; 16/16 in restoration
Held-out	sealed

8. Final route-level results

Table 2 reports family-level matched increments after averaging the two prompt orders. The extension priority is the frozen selection order and is not an evidence ranking. All values are logits. 'Sign match' is the fraction of 89 families for which noising and restoration matched increments have the same sign.

P	Tested route	Noise mean Delta	Noise screen	Restore mean Delta	Restore screen	Sign match
1	L15H25 query_sentence -> [L16H1, L16H21] -> L18H18 Q	+0.217	coherent + heterogeneous	+0.222	coherent + heterogeneous	96.6%
2	L17H1 query_writer_union -> [L18H18, L18H19] -> L27H6 Q	+1.635	coherent + heterogeneous	+1.511	coherent + heterogeneous	98.9%
3	L20H7 colon -> MLP20+block21+block22 -> logits	-0.069	heterogeneous only	-0.061	heterogeneous only	47.2%
4	L26H23 colon -> MLP26+block27+block28 -> logits	-0.183	coherent + heterogeneous	-0.166	coherent + heterogeneous	91.0%
5	L8H15 query_sentence -> MLP8+MLP9 -> L18H18 V	-0.160	coherent + heterogeneous	-0.154	coherent + heterogeneous	100.0%
6	L17H1 query_writer_union -> [L18H18] -> L27H6 Q	+0.876	coherent + heterogeneous	+0.809	coherent + heterogeneous	96.6%
7	L17H1 query_writer_union -> [L18H19] -> L27H6 Q	+0.430	coherent + heterogeneous	+0.409	coherent + heterogeneous	95.5%
8	L17H1 query_writer_union -> [L18H18, L18H19] -> L21H6 Q	+0.208	coherent + heterogeneous	+0.194	coherent + heterogeneous	97.8%
9	L17H1 query_writer_union -> [L18H18, L18H19] -> L22H5 Q	+0.194	coherent + heterogeneous	+0.170	coherent + heterogeneous	96.6%
10	L15H25 query_child_last -> [L16H1, L16H21] -> L18H18 Q	+0.172	coherent + heterogeneous	+0.179	coherent + heterogeneous	96.6%

P	Tested route	Noise mean Delta	Noise screen	Restore mean Delta	Restore screen	Sign match
11	L17H1 query_writer_uni on -> [L18H18, L18H19] -> L21H18 Q	+0.159	coherent + heterogeneous	+0.174	coherent + heterogeneous	97.8%
12	L8H15 query_sentence -> MLP9 -> L18H18 V	-0.145	coherent + heterogeneous	-0.140	coherent + heterogeneous	100.0%
13	L8H15 query_sentence -> MLP9 -> L18H19 V	-0.147	coherent + heterogeneous	-0.147	coherent + heterogeneous	100.0%
14	L15H25 query_sentence -> [L16H1, L16H21] -> L18H19 Q	+0.151	coherent + heterogeneous	+0.149	coherent + heterogeneous	96.6%
15	L26H23 colon -> block27 -> logits	-0.089	heterogeneous only	-0.084	heterogeneous only	86.5%
16	L8H15 query_sentence -> MLP8+MLP9 -> L18H19 V	-0.131	coherent + heterogeneous	-0.133	coherent + heterogeneous	97.8%

9. Interpretation by source / chain

9.1 L17H1 -> layer-18 pair -> later Q receivers

The largest matched effect is the joint L18H18+L18H19 release into L27H6 Q: noise +1.635 and restoration +1.511, with 98.9% family-level noising/restoration sign agreement. L18H18 alone (+0.876 / +0.809) and L18H19 alone (+0.430 / +0.409) are also coherent at L27H6. The joint pair is additionally coherent into L21H6 (+0.208 / +0.194), L22H5 (+0.194 / +0.170), and L21H18 (+0.159 / +0.174).

These results support the proposed ordered L17H1 -> layer-18 -> later-Q composition under the selective-live-head intervention. They do not establish that the heads encode a literal relation, that the measured routes are unique or minimal, or that the joint effect can be decomposed additively into the two single-head effects.

9.2 L15H25 -> L16H1/L16H21 -> layer-18 Q

All three selected chains remain coherent in both directions. The query-sentence route through the joint L16H1+L16H21 release into L18H18 Q is +0.217 in noising and +0.222 in restoration. The child-last route into L18H18 is +0.172 / +0.179, and the query-sentence route into L18H19 is +0.151 / +0.149. Family-level directional sign agreement is 96.6% for each selected route.

This supports composition of the tested L15 -> L16 -> L18 paths in the discovery population. It does not establish individual necessity of both L16 heads because the full extension used the prospectively selected joint releases.

9.3 L8H15 -> MLP9-associated processing -> layer-18 V

All four selected L8H15 routes remain coherent and negative in both directions. The mean matched increments range from -0.160 to -0.131 in noising and -0.154 to -0.133 in restoration. The MLP9-only routes have 100% family sign agreement within each direction and 100% noising/restoration family sign match. MLP8+MLP9 is similar in magnitude.

The supported interpretation is an opposing or attenuating intervention dependency involving MLP9 and L18 V under the tested hybrid. A negative matched increment is not evidence for a 'negative representation' or negative information.

9.4 L26H23 downstream block-mediated routes

The full release MLP26+block27+block28 remains coherent: -0.183 in noising and -0.166 in restoration, with 91.0% family-level directional sign agreement. The block27-only route is weaker: -0.089 / -0.084. It remains retained only through the heterogeneous rule, although its noising/restoration family sign match is 86.5%.

The full release therefore provides the clearer block-mediated result. The block27-only condition should be reported as heterogeneous rather than coherent at full-discovery scale. The experiment does not identify which constituent heads inside the released blocks are responsible.

9.5 L20H7 downstream block-mediated route

The selected full release MLP20+block21+block22 has aggregate means -0.069 in noising and -0.061 in restoration. It is retained only by the heterogeneous rule in both directions. More importantly, only 47.2% of families have matching noising/restoration signs, and the descriptive family-level correlation is -0.527.

Accordingly, L20H7 should not be described as a stable directed negative route. The result is better characterized as heterogeneous, cancellation-prone downstream sensitivity whose aggregate mean is negative in both directions but whose family-level behavior differs substantially between the two interventions.

9.6 Negative and unextended findings

The full extension does not change the initial-panel negative findings for L13H18 and L16H31 because no route from either source passed the frozen initial screen and neither entered the extension. Likewise, the six initially retained configurations omitted by the 16-route cap remain 20-family-only results. Their absence from the extension is a scope boundary, not evidence against them.

10. Stability, selection-excluded families, and reproducibility

All 16 selected routes preserve the same aggregate mean sign when the noising analysis expands from 20 to all 89 discovery families, and all 16 remain retained in both directions. Because the same 20 families were used for selection, this full-89 result is not an independent validation.

As an additional CPU-only sensitivity check, the 69 discovery families that were not part of the selection panel were analyzed separately without changing the route set or thresholds. The qualitative classification is unchanged: 16/16 routes are retained in both directions, 14/16 are coherent in both, and the same two routes - L20H7 full release and L26H23 block27-only - are heterogeneous-only. This is still discovery-split evidence, not held-out validation.

69-family sensitivity	Noise	Restore	Sign match	r	Classification
L20H7 full release	-0.059	-0.069	43.5%	-0.575	heterogeneous only
L26H23 block27 only	-0.078	-0.073	87.0%	+0.831	heterogeneous only

The common-20 reproducibility check is numerically exact to floating-point precision: the maximum differences in mean signed effect and mean absolute effect are on the order of 1e-16, with identical sign-agreement, active-fraction, and retention labels. This is an implementation reproducibility check, not an independent scientific replication.

11. Prompt-order sensitivity

Family averaging should not be interpreted as order invariance. The largest L17H1 -> {L18H18,L18H19} -> L27H6 Q route shows a substantial magnitude difference between the two prompt orders: noising means are +2.220 versus +1.050, and restoration means are +2.085 versus +0.938. The effect remains

positive in aggregate and across almost all families, but its magnitude depends on prompt order. Similar, smaller order sensitivity occurs in several other L17 routes.

12. Figures

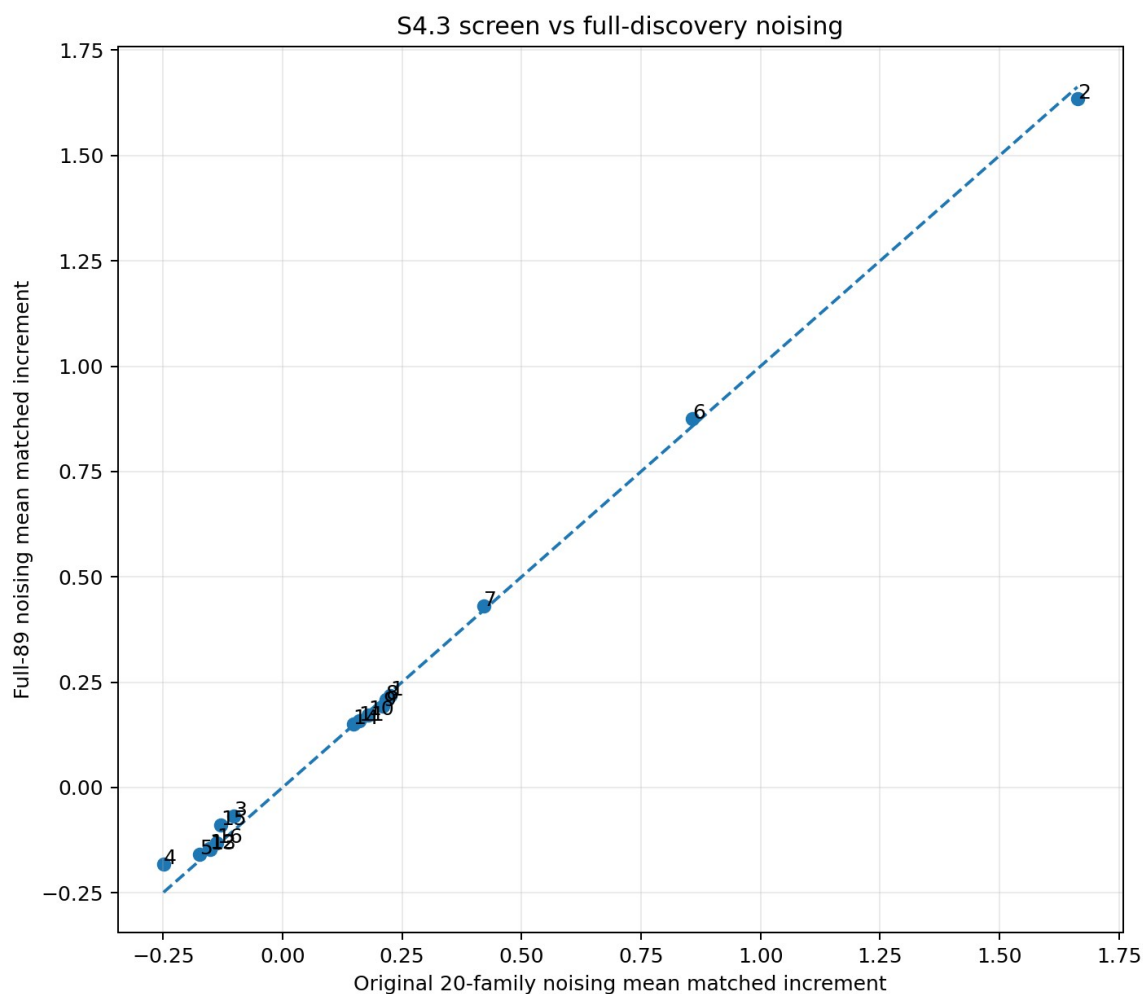


Figure 1. Original 20-family noising mean matched increments versus full-89 noising means for the 16 prospectively selected routes. This is a stability comparison, not independent validation.

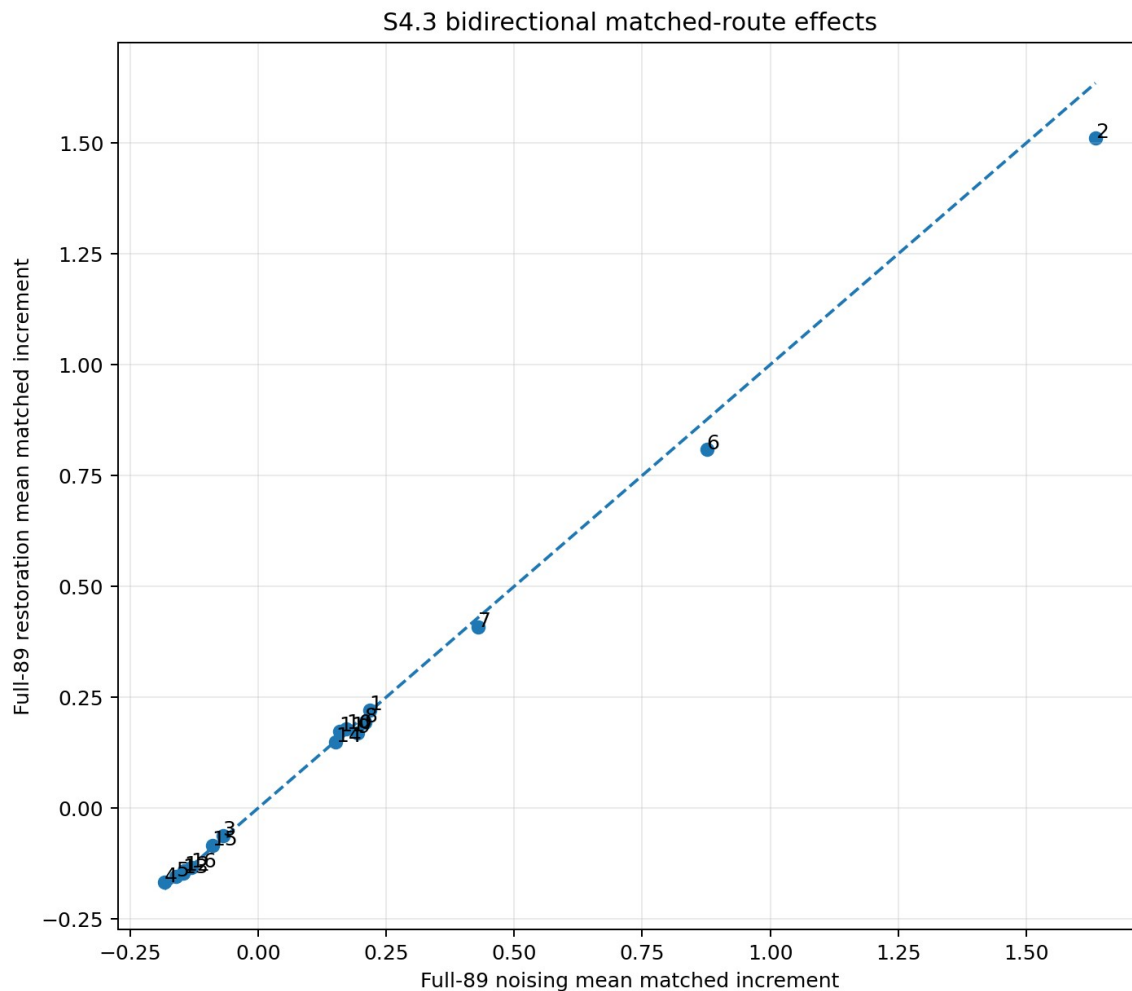


Figure 2. Full-89 noising versus reverse-restoration mean matched increments. Aggregate signs agree for all 16 selected routes, while family-level agreement differs by route.

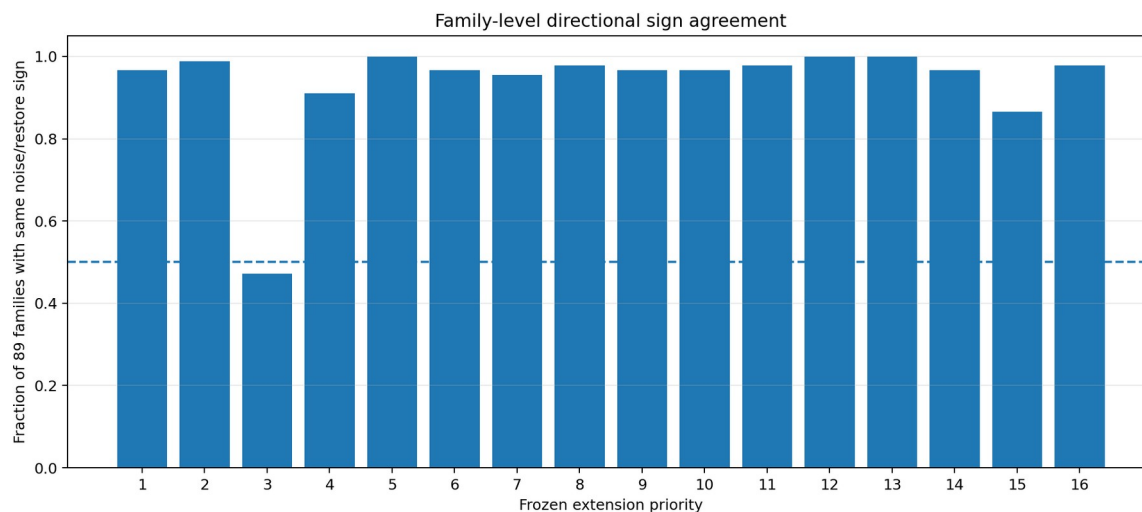


Figure 3. Fraction of discovery families with the same noising/restoration effect sign. L20H7 is the clear low-agreement case.

13. What S4.3 supports - and what it does not

Supported, within the tested discovery population and intervention semantics:

- Specific composed dependencies from L17H1 through L18H18/L18H19 into several later Q receivers, with the largest effect at L27H6.

- Smaller but consistent composed dependencies from L15H25 through the joint L16H1/L16H21 release into L18H18/L18H19 Q.
- Consistent opposing/attenuating L8H15 dependencies involving MLP9-associated processing into L18 V.
- A downstream block-mediated L26H23 effect, with the full release more stable than block27 alone.
- A heterogeneous L20H7 downstream sensitivity that should remain explicitly qualified.

Not established by S4.3:

- A unique, minimal, or complete circuit.
- A fraction of model reasoning, causal information, or mediation attributable to a route.
- Literal semantic content such as a 'mother representation' or 'query-selection code'.
- Individual-head attribution from whole-block release.
- Irrelevance of untested routes or of initially retained configurations omitted by the extension cap.
- Held-out generalization or universal behavior beyond the discovery population.

14. Final S4.3 conclusion and next boundary

The bounded S4.3 plan has been executed to completion. The full-discovery extension supports the main L17H1 and L15H25 composed dependencies and the L8H15 opposing dependency under the frozen interventions. It also narrows the interpretation of the downstream block results: the full L26H23 release remains coherent, block27 alone is heterogeneous-only, and L20H7 is heterogeneous and directionally unstable at the family level.

No additional S4.3 search is warranted by the frozen protocol. Any next stage should keep the route set and interpretation fixed before accessing held-out families. S4.4/S4.5 mechanism/group tests can use these S4.3 routes as explanatory context, but a faithful retained head set need not imply that every incoming local route has been identified.

Appendix A. Reproducibility identifiers

Artifact	Identifier
Production stage	S4.3_extension_v3_family034_compat_hardware_independent
Production identity hash	1f6d18be982d12ce7d1c0a4a88f19f2fe94bff4c5af89ee66c6bb46a2836039b
Post-extension analysis	S4.3_post_extension_full89_bidirectional_v1
Analysis manifest SHA-256	7833704dcda0ff85e78b58bbcd7a7eac1211f6810fa15ae588204ea230d33007
Initial analysis manifest SHA-256	f7929c84ef8a0317c0052644a94029b1c70ee5b28b2b838d85949defd1a7dda9
Selected contrasts SHA-256	0ae04006e7757ef987023d925d4ea3d936c7ffddcea209bb64cd897b80375413
Selected prerequisites SHA-256	3f4c9ada32d01cc76884b20771dc606a17cbd6b59e44833f8acfc8e12bba5d87
Held-out	sealed